Probability model to solve the school bus routing problem with stops selection

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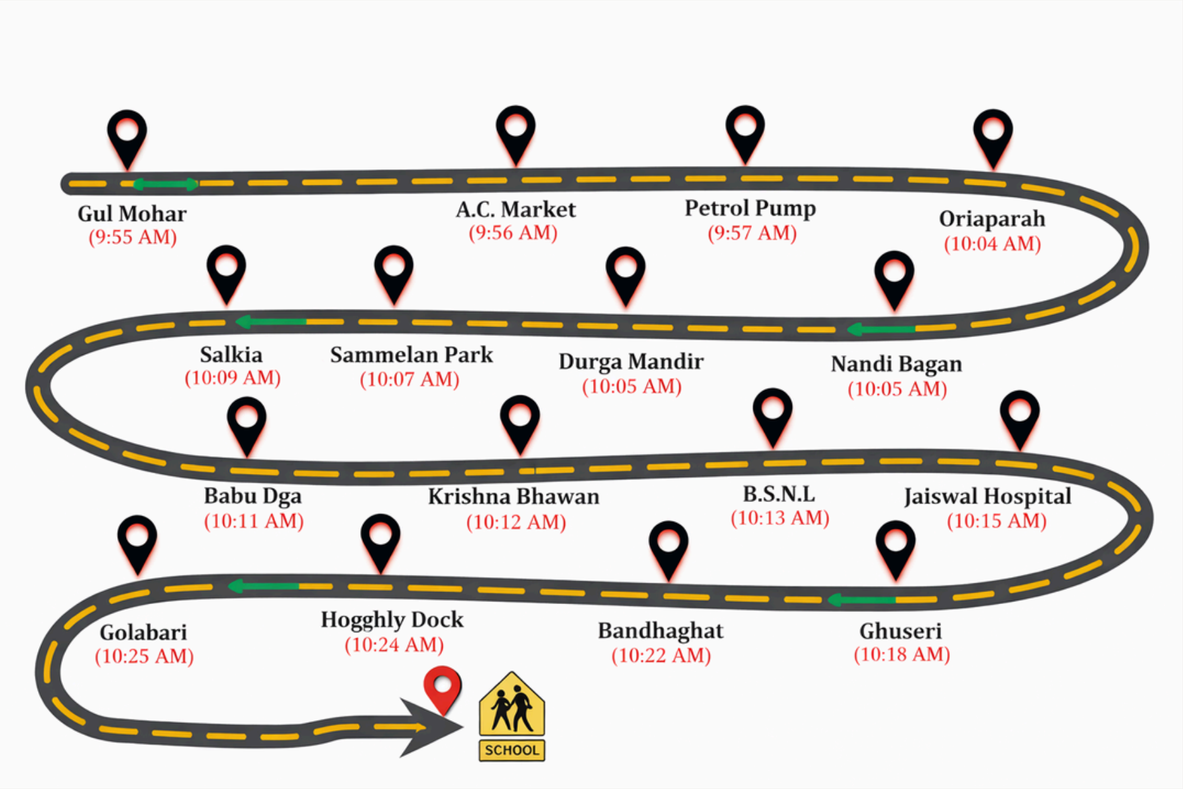


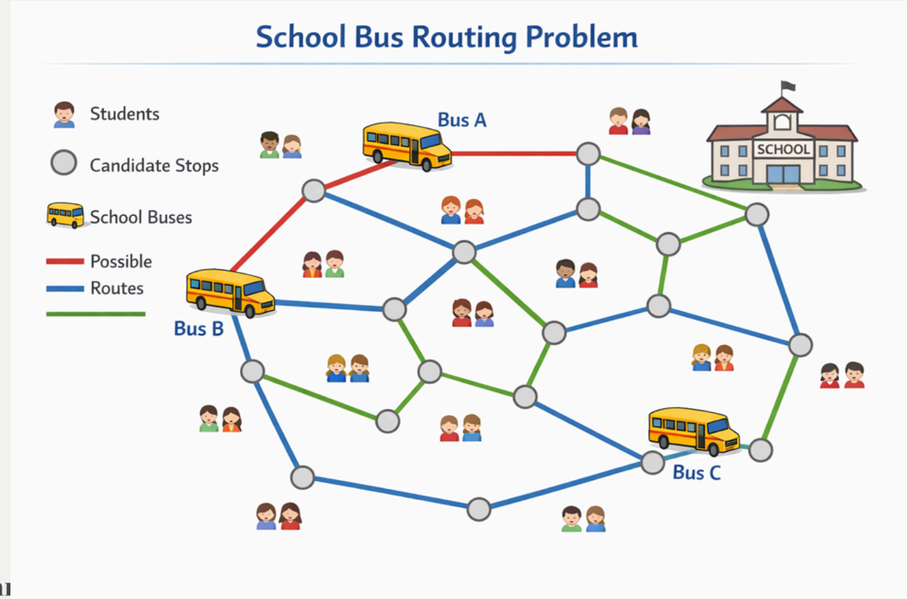
Made by:
Saumya Shah(AU2440274) &
Pratyusha Sharma (AU2440012)

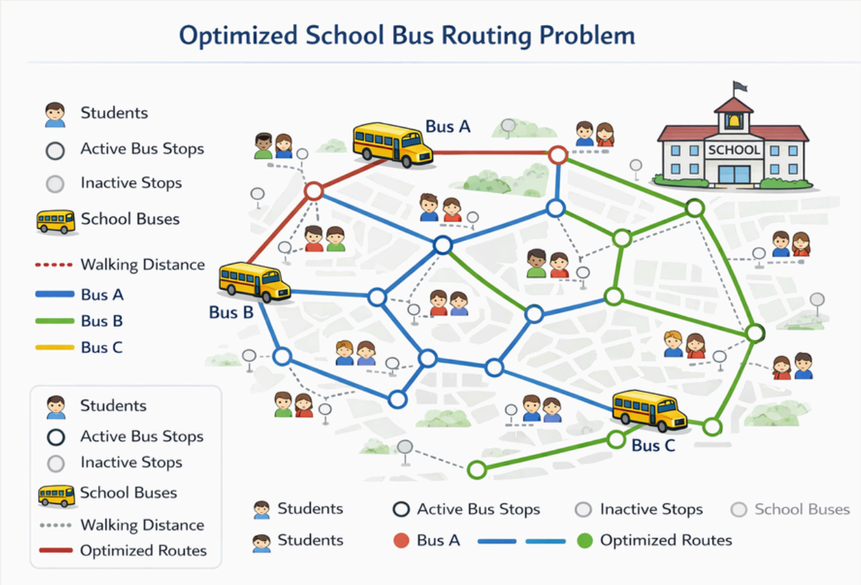
Project Context and Motivation

- **Application Domain:** Intelligent Transportation System (ITS) [school bus - routing]

- **Real-World Context:** Student transport with changing stops and constrsints

- **Reason for uncertainty:** Combinational routing choices and interacting constraints





Project Objective and Scope

Exact Objective:

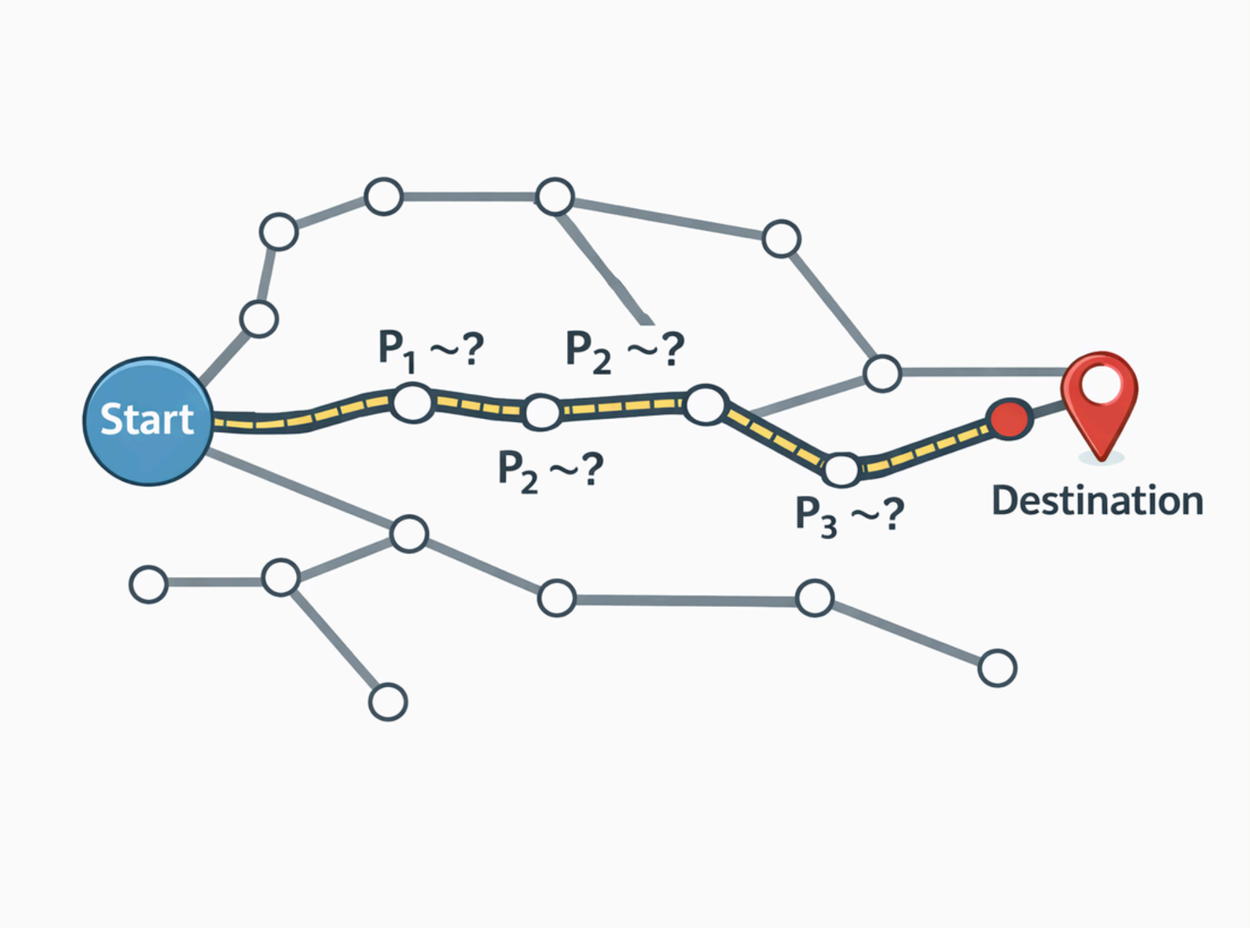
- Minimize expected travel time ofall school buses

Decision Task:

- Select stops, asign students and sequence routes

Scope:

- Static probabilistic road network



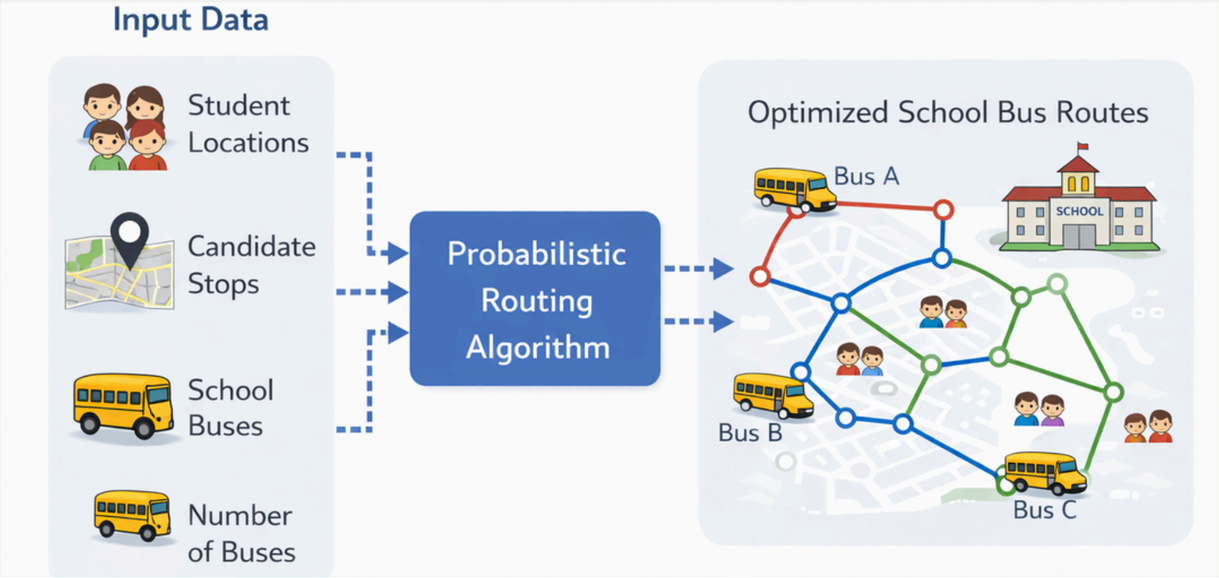
Project System Overview

- **Pipeline:**

Input data → ***Allocation*** → ***probabilistic routing*** → ***Evaluation*** → ***Probability Modeling*** → ***Sampling***

- **Components:** Student-stop assignment,EDA- based routing

- **Data Flow:** Fitness guides probability updates



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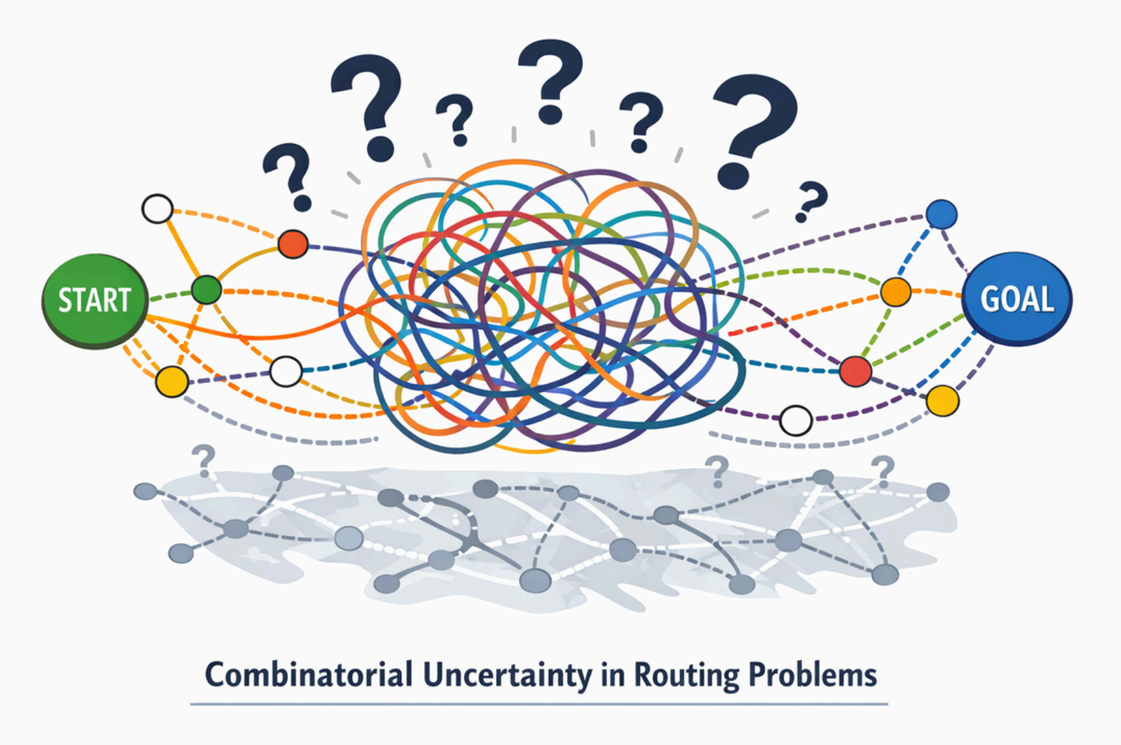
Source of Uncertainty in the Project

- **Measurement Uncertainty:** Approximate student and stop locations
- **Contextual Variability:** Changing student distributions and constraints
- **Modeling Uncertainty:** Multiple valid route permutations



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Key Random Variables and Their Roles

- **Primary Random variable:** Positions of bus stops in route permutations

- **Representation:** Probability of assigning a stop toa route problem

Probabilistic Models and Assumptions

- **Model used:** Estimation of Distribution Algorithm
- **Assumption:** Independence between route positions
- **Justification:** Combinational routing choices and interacting constraints

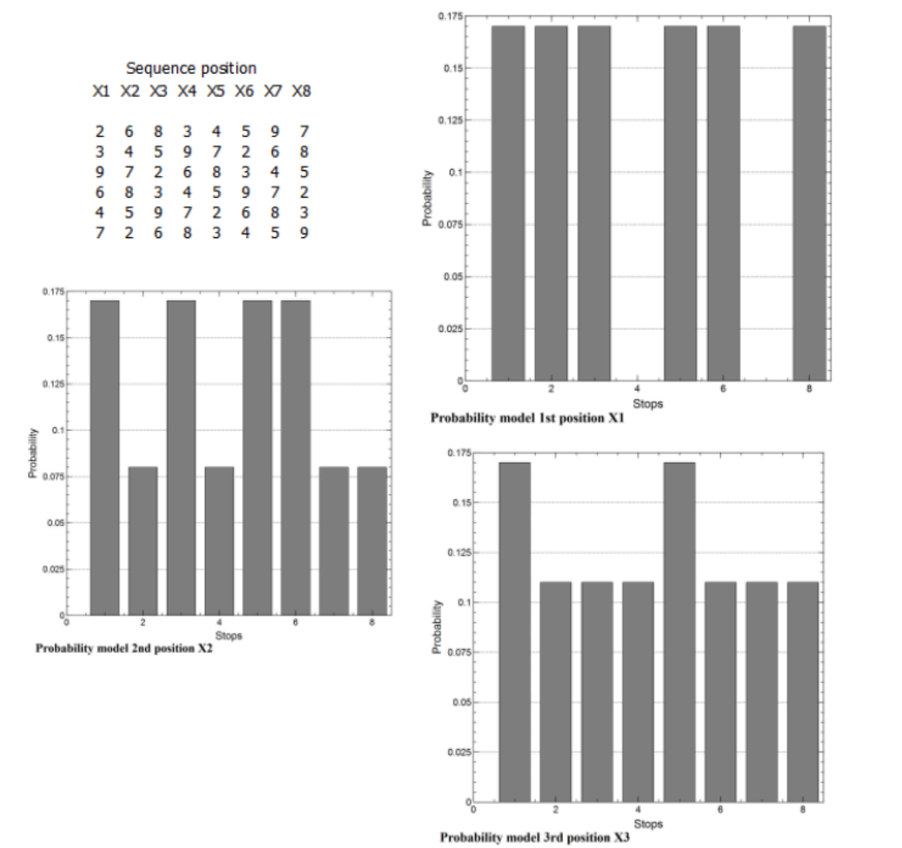

Fig. 2. Probability distribution generated for each bus stop.

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Probabilistic Reasoning and Inference Logic

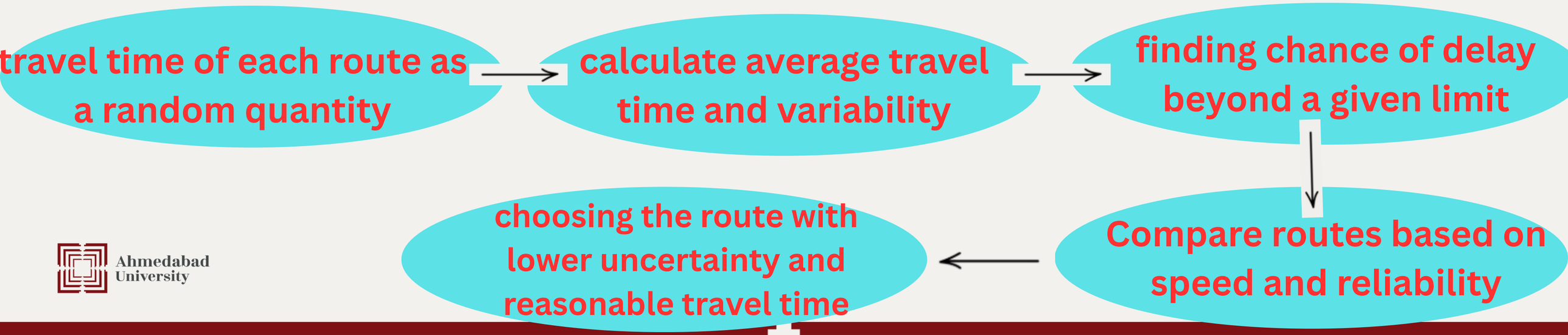
Uncertainty Propagation

- Large combinations of stop selections and route permutations create combinatorial uncertainty.
- Student allocation and capacity constraints affect feasibility, while probability distributions progressively reduce uncertainty across generations.

Role of Probabilistic Reasoning:

- Estimate marginal probability distributions from elite solutions to capture patterns of high-quality routes.
- Use these learned distributions to guide probabilistic sampling and iteratively improve solution quality.

Conceptual Reasoning Pathway:



Operationalizing Probabilistic Reasoning

Slide 7 Reasoning

- Learn marginal probability distributions from elite routes.
- Generate improved routes through probabilistic sampling and iteration.

Evaluation Metrics & Probabilistic Interpretation

- Minimize total travel time as fitness measure.
- Population statistics reflect learned route structure.

Simplifying Assumptions

- Independence between route positions.
- Fixed capacity and walking distance.

Current Limitations and Gaps

Known Weakness

- Independence assumption may ignore dependencies between stop positions.
- Performance may decline for very large-scale instances.

Unverified / Questionable Assumptions

- Travel times treated as deterministic (no traffic uncertainty).
- Student allocation fixed before routing optimization.

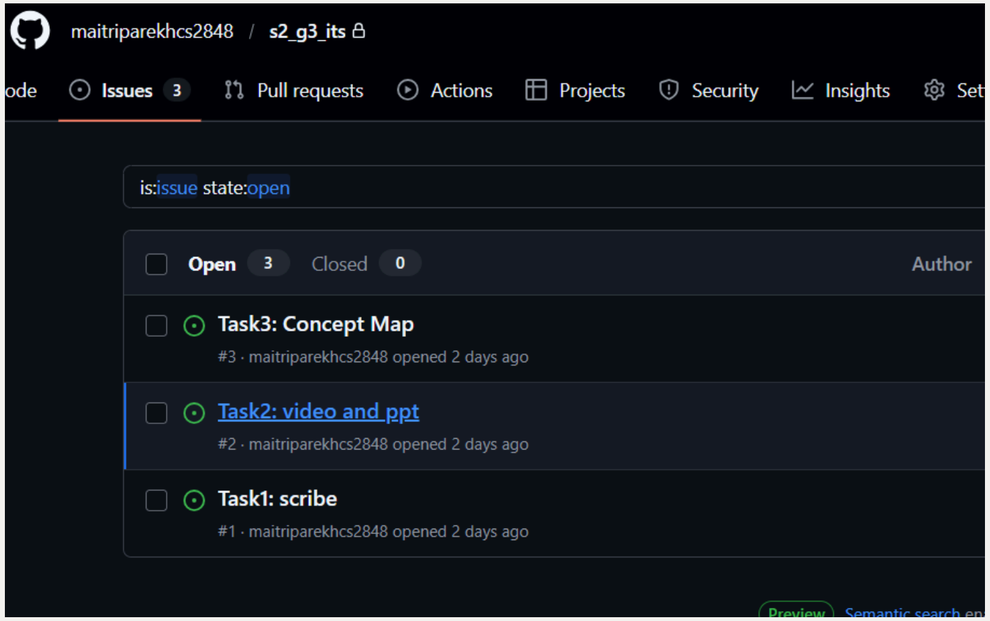
Conceptual Ambiguities

- Interaction between stop selection and routing not fully modeled jointly.
- Effect of population size and parameters on convergence not deeply analyzed.

Planned Refinements and Role Coordination

What Will Evolve in the Next Milestone?

- Improve probability model and experiment with parameter tuning.
- Expand testing and refine route optimization performance.



Conceptual / Modeling Tasks Ahead

- Explore modeling dependencies between route positions.
- Analyze scalability and convergence behavior.

Role of Reviewer / GitHub Manager

- Create repository and add team members as collaborators.
- Assign issues, review contributions, and approve pull requests.

Summary of Group-Level Understanding

- The routing problem is modeled as a probability distribution over feasible route permutations.
- Estimation of Distribution Algorithm (EDA) learns structural patterns from elite solutions instead of using random evolutionary operators.
- Marginal probability distributions guide solution generation and progressively reduce uncertainty.
- Probabilistic modeling enables systematic exploration of a large combinatorial search space.

Reference

Pérez-Rodríguez, R., & Hernández-Aguirre, A. (2016). Probability model to solve the school bus routing problem with stops selection. International Journal of Combinatorial Optimization Problems and Informatics, 7(1), 30-39